

- **Module 14) Python – Collections, functions and Modules**

8. Functions

- **Defining functions in Python**

A **function** in Python is a reusable block of code that performs a specific task. Functions help organize code, avoid repetition, and improve readability.

Basic Structure of a Function

```
def function_name(parameters):  
    # function body  
    return result
```

Key Components

Component	Description
def keyword	Used to define a function
function_name	The name used to call the function
parameters	Optional variables passed into the function (also called <i>arguments</i>)
:	Indicates the start of the function block
Function body	The indented block of code that runs when the function is called

Component	Description
return	Optional keyword to send back a result or value to the caller

Types of Functions

1. **Built-in Functions:** Provided by Python (len(), print()).
2. **User-defined Functions:** Created using def.
3. **Lambda Functions:** Anonymous, one-line functions using the lambda keyword.

- **Different types of functions: with/without parameters, with/without return values.**

Functions in Python can be categorized into four main types depending on:

1. Whether they **accept parameters**
2. Whether they **return a value**

1. Functions Without Parameters and Without Return Values

- **Purpose:** Perform a task without needing input or returning a result.
- **Structure:**

```
def greet():
    print("Hello!")
```

2. Functions With Parameters and Without Return Values

- **Purpose:** Accept input, process it, but don't return anything.

- **Structure:**

```
def greet(name):  
    print("Hello,", name)
```

3. Functions Without Parameters and With Return Values

- **Purpose:** Don't need input, but generate and return a result.
- **Structure:**

```
def get_pi():  
    return 3.14159
```

4. Functions With Parameters and With Return Values

- **Purpose:** Accept input and produce a result based on that input.
- **Structure:**

```
def add(a, b):  
    return a + b
```

- **Anonymous functions (lambda functions).**

An **anonymous function** is a function **without a name**.

In Python, these are created using the **lambda** keyword and are commonly referred to as **lambda functions**.

Syntax of Lambda Functions:

lambda arguments: expression

- **lambda:** Keyword to define the anonymous function.
- **arguments:** Input values (like parameters in regular functions).
- **expression:** A single expression whose result is returned automatically.

Characteristics of Lambda Functions

Feature	Description
No function name	Defined without def and no name is given.
Single-expression body	Can only contain one expression (no statements or loops).
Returns value	Automatically returns the result of the expression.
Often short-lived	Typically used for short operations, often inline.